Ps 02

Erik An

<2026-03-16 Mon>

Contents

1 Problem 1.24 DONE 2

1.1 Task 2

1.2 Solution 2

2 Problem 1.29 DONE 3

2.1 Task 3

2.2 Solution 3

3 Problem 1.33 DONE 3

3.1 Task 3

3.2 Solution 3

4 Problem 1.37 DONE 5

4.1 Task 5

4.2 Solution 5

4.2.1 (1) 5

4.2.2 (2) 5

1 Problem 1.24 DONE

1.1 Task

(Reverse Triangle Inequality). From the triangle inequality, derive the *reverse triangle inequality*, namely

$|\|x\| - \|y\|| \leq \|x - y\|, \forall (x, y) \in \mathbb{K}$

where $\mathbb{K} = \mathbb{R}^n$ or $\mathbb{C}^n$ and $\|\cdot\|$ is any norm in $\mathbb{R}^n$.

1.2 Solution

First, rewrite $x = (x - y) + y$ and apply the triangle inequality:

$$\begin{aligned} \|x\| &= \|(x - y) + y\| \\ &\leq \|x - y\| + \|y\| \end{aligned}$$

which gives:

$$\|x\| - \|y\| \leq \|x - y\|$$

Similarly, rewrite $y = (y - x) + x$ and apply the triangle inequality:

$$\begin{aligned} \|y\| &= \|(y - x) + x\| \\ &\leq \|y - x\| + \|x\| \end{aligned}$$

By homogeneity, $\|y - x\| = \|-(x - y)\| = |-1| \cdot \|x - y\| = \|x - y\|$, so:

$$\|y\| - \|x\| \leq \|x - y\|$$

Combining (1) and (2): since both $\|x\| - \|y\| \leq \|x - y\|$ and $-(\|x\| - \|y\|) \leq \|x - y\|$, we have:

$$\big|\|x\| - \|y\|\big| \leq \|x - y\|$$

2 Problem 1.29 DONE

2.1 Task

(Vector space) On $\mathbb{R}^2$ we define the following operations
$(x_1, x_2) \oplus (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$, $\lambda \odot (x_1, x_2) = (\lambda x_1, 0)$.
Which vector space axioms does $(\mathbb{R}^2, \oplus, \odot)$ satisfy? Is it a vector space?

2.2 Solution

1. **Associativity of addition** $(x \oplus y) \oplus z = (x_1 + y_1 + z_1,\ x_2 + y_2 + z_2) = x \oplus (y \oplus z)$.
2. **Commutativity of addition** $x \oplus y = (x_1 + y_1,\ x_2 + y_2) = y \oplus x$.
3. **Additive identity** $(x_1, x_2) \oplus (0, 0) = (x_1, x_2)$.
4. **Additive inverse** $(x_1, x_2) \oplus (-x_1, -x_2) = (0, 0)$.
5. **Distributivity over scalar addition** Both sides yield $((\lambda + \mu)x_1,\ 0)$.
6. **Distributivity over vector addition** Both sides yield $(\lambda(x_1 + y_1),\ 0)$.
7. **Compatibility of scalar multiplication** Both sides yield $(\lambda\mu\, x_1,\ 0)$.
8. **Scalar multiplicative identity** $1 \odot (x_1, x_2) = (x_1, 0) \neq (x_1, x_2)$ when $x_2 \neq 0$.

Conclusion: $(\mathbb{R}^2, \oplus, \odot)$ is not a vector space.

Axioms 1-7 hold, but axiom 8 fails: scalar multiplication by 1 zeroes out the second component.

3 Problem 1.33 DONE

3.1 Task

(Bases and regular matrices) Let $\mathcal{U} = \{u_1, \ldots, u_n\}$ be a basis of a vector space $\mathbb{E}$. Let $A \in \mathbb{K}^{n \times n}$ and vectors $v_j = \sum_{i=1}^{n} A_{ij} u_i$. Prove that $\mathcal{V} = \{v_1, \ldots, v_n\}$ is also a basis if and only if A is invertible.

3.2 Solution

- $(\Rightarrow)$
  If $\mathcal{V}$ is basis, every u_i can be expressed as a linear combination of v_j:

$$u_i = \sum_{j=1}^{n} B_{ji} v_j$$

for some matrix $B \in \mathbb{K}^{n \times n}$, where B_{ji} is the coefficient of v_j in the expansion of u_i.

Now substitute $v_j = \sum_{k=1}^{n} A_{kj} u_k$:

$$u_i = \sum_{j=1}^{n} B_{ji} \sum_{k=1}^{n} A_{kj} u_k = \sum_{k=1}^{n} \left(\sum_{j=1}^{n} A_{kj} B_{ji} \right) u_k$$

The inner sum is exactly the (k, i) entry product AB.

$$u_i = \sum_{k=1}^{n} (AB)_{ki} \, u_k$$

Since $\mathcal{U}$ is a basis, the representation of u_i in terms of $u_1, \cdots, u_n$ is unique, and it is simply $u_i = 0 \cdot u_1 + \cdots + 1 \cdot u_i + \cdots + 0 \cdot u_n$. Therefore,

$$(AB)_{ki} = \delta_{ki} = \begin{cases} 1 & k = i \\ 0 & k \neq i \end{cases}$$

This gives $AB = I$. Since A is a square matrix with a right inverse, it is invertible.

- ($\Leftarrow$)

  If A is invertible, we need to show $\mathcal{V}$ is a basis. Since $|\mathcal{V}| = n = \dim \mathbb{E}$, it suffices to show $\mathcal{V}$ is linearly independent.

  Suppose $\sum_{j=1}^{n} x_j v_j = 0$ for some scalars $x_1, \ldots, x_n$.

  Substitute $v_j = \sum_{i=1}^{n} A_{ij} u_i$:

$$\sum_{j=1}^{n} x_j \sum_{i=1}^{n} A_{ij} u_i = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} A_{ij} x_j \right) u_i = \sum_{i=1}^{n} (Ax)_i \, u_i = 0$$

  Since $\mathcal{U}$ is a basis, the u_i are linearly independent, so every coefficient must vanish:

$$(Ax)_i = 0 \quad \text{for all } i = 1, \ldots, n$$

  This means $Ax = 0$. Since A is invertible, $x = A^{-1} \cdot 0 = 0$, i.e. $x_1 = \cdots = x_n = 0$.

  Therefore $\mathcal{V}$ is linearly independent, and since it has n vectors in an n dimensional space, $\mathcal{V}$ is a basis.

4 Problem 1.37 DONE

4.1 Task

(Direct sum of triangular matrices) Let $\mathbb{E} = \mathbb{K}^{n\times n}$, we define $\mathbb{F}_1, \mathbb{F}_2 \subset \mathbb{E}$ as the subspaces of upper and lower triangular matrices, respectively. Determine whether $\mathbb{E} = \mathbb{F}_1 \oplus \mathbb{F}_2$. Does the result vary if we replace $\mathbb{F}_1$ by $\mathbb{F}_1'$ the subspace of 1-stricly triangular matrices (as defined in ex. 1.28)?

4.2 Solution

4.2.1 (1)

Proof.

We determine whether $\mathbb{E} = \mathbb{F}_1 \oplus \mathbb{F}_2$ using the Dimension Theorem for Sums of Subspaces:

$$\dim(\mathbb{F}_1 + \mathbb{F}_2) = \dim(\mathbb{F}_1) + \dim(\mathbb{F}_2) - \dim(\mathbb{F}_1 \cap \mathbb{F}_2).$$

A matrix that is simultaneously upper and lower triangular can only have nonzero entries on the diagonal, so

$\mathbb{F}_1 \cap \mathbb{F}_2 = \mathcal{D}_n$ and $\dim(\mathbb{F}_1 \cap \mathbb{F}_2) = n$.

Plugging in,

$$\begin{aligned}\dim(\mathbb{F}_1 + \mathbb{F}_2) &= \frac{n(n+1)}{2} + \frac{n(n+1)}{2} - n \\ &= n^2 + n - n \\ &= n^2 = \dim(\mathbb{E}).\end{aligned}$$

So $\mathbb{F}_1 + \mathbb{F}_2 = \mathbb{E}$. However, for the sum to be direct we require $\dim(\mathbb{F}_1 \cap \mathbb{F}_2) = 0$, and we found it equals $n \geq 1$.

Therefore $\mathbb{E} \neq \mathbb{F}_1 \oplus \mathbb{F}_2$.

4.2.2 (2)

Proof.

We determine whether $\mathbb{E} = \mathbb{F}_1' \oplus \mathbb{F}_2$ by checking the two conditions for a direct sum.

A matrix $A \in \mathbb{F}_1' \cap \mathbb{F}_2$ satisfies

$a_{ij} = 0$ for $i \geq j$ (strictly upper triangular) and

$a_{ij} = 0$ for $i < j$ (lower triangular). Together these force

$a_{ij} = 0$ for all i, j, so

$\mathbb{F}_1' \cap \mathbb{F}_2 = \{0\}$.

It remains to verify that $\mathbb{F}_1' + \mathbb{F}_2 = \mathbb{E}$. By the Dimension Theorem for Sums of Subspaces,

$$\begin{aligned}
\dim(\mathbb{F}_1' + \mathbb{F}_2) &= \dim(\mathbb{F}_1') + \dim(\mathbb{F}_2) - \dim(\mathbb{F}_1' \cap \mathbb{F}_2) \\
&= \frac{n(n-1)}{2} + \frac{n(n+1)}{2} - 0 \\
&= \frac{n^2 - n + n^2 + n}{2} \\
&= n^2 = \dim(\mathbb{E}).
\end{aligned}$$

Both conditions are satisfied, so $\mathbb{E} = \mathbb{F}_1' \oplus \mathbb{F}_2$.